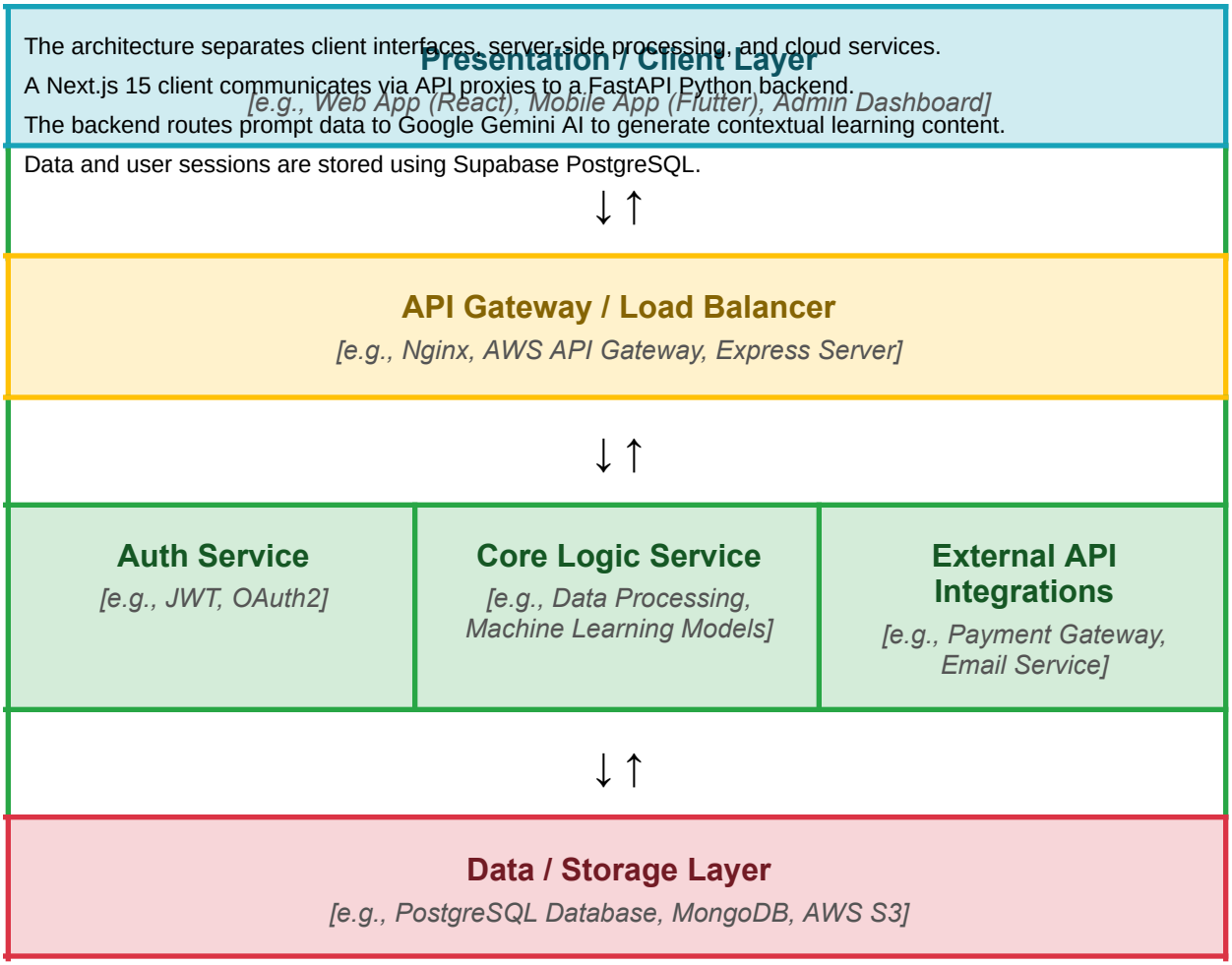
Solution Architecture

Date	02 October 2026
Team ID	SWTID-2026-1529
Project Name	EduGenie: Google gemini powered learning assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer] Next.js 15 App	[Describe what this component does] React 19 Frontend with Glassmorphic UI & Speech synthesis.	[List specific tech]
[API Gateway] FastAPI Backend	[Describe what this component does] Python 3.14 server for prompt pipeline & JSON validation.	[List specific tech]
[Microservices] Gemini AI Engine	[Describe what this component does] Generates Smart Q&A, MCQs, Summaries, & Roadmaps.	[List specific tech]
[Database] Supabase DB	[Describe what this component does] Handles user auth, session logging, & study session caching.	[List specific tech]